

End-of-Studies Internship Report

Program: Artificial Intelligence and Computer Engineering

• • Safae Nagbi • •

Title

*Agricultural Intelligence:
Crop Mapping using Machine Learning
and Sentinel-2 Satellite Imagery*

Host Company: Oracle Morocco Research and Development Center

Professional Supervisor: Mitra Noosha

Academic Supervisor: Soukaina Bouhsissin

From 15/01/2026 **to** 14/07/2026

Academic Year: 2025–2026

*To my beloved family,
for their unconditional love, endless patience,
and constant encouragement throughout my journey.*

*To my dear friends,
for their support, motivation, and presence
during both the challenging and joyful moments.*

*This work is dedicated to all those
who believed in me and helped me move forward.*

— Safae Nagbi

Acknowledgments

I would like to express my sincere gratitude to all those who contributed, directly or indirectly, to the successful completion of this internship and the preparation of this report.

First, I would like to thank Oracle Corporation and the Oracle Morocco Research and Development Center for welcoming me and giving me the opportunity to carry out this internship in a professional and stimulating environment. This experience allowed me to strengthen my technical skills, discover real-world industrial practices, and better understand the challenges related to digital government and data-driven solutions.

I would like to express my special thanks to my internship mentor, Mitra Noosha, for her guidance, availability, valuable advice, and continuous support throughout the project. Her supervision helped me better understand the project objectives, overcome technical challenges, and progress with confidence during the different phases of the internship.

I am also deeply grateful to the members of my team, especially Sam Batterman, ElQajjam Mohammed, Hamza EL MAKRINI, Lahoucine Digrari, Anass Zouaiga, Sara Essalhi, Imane Jabal, and all the other team members, for their warm welcome, collaboration, support, and willingness to share their knowledge and experience.

I would like to thank my academic mentor, Soukaina Bouhsissin, as well as the ENSAM pedagogical team and all my professors for their guidance, support, and the knowledge they have provided throughout my academic journey.

Finally, I would like to express my deepest appreciation to my family and friends for their constant encouragement, patience, and support. Their presence and motivation have been invaluable throughout this experience.



Résumé

Résumé : Nulla malesuada porttitor diam. Donec felis erat, congue non, volutpat at, tincidunt tristique, libero. Vivamus viverra fermentum felis. Donec nonummy pellentesque ante. Phasellus adipiscing semper elit. Proin fermentum massa ac quam. Sed diam turpis, molestie vitae, placerat a, molestie nec, leo. Maecenas lacinia. Nam ipsum ligula, eleifend at, accumsan nec, suscipit a, ipsum. Morbi blandit ligula feugiat magna. Nunc eleifend consequat lorem. Sed lacinia nulla vitae enim. Pellentesque tincidunt purus vel magna. Integer non enim. Praesent euismod nunc eu purus. Donec bibendum quam in tellus. Nullam cursus pulvinar lectus. Donec et mi. Nam vulputate metus eu enim. Vestibulum pellentesque felis eu massa.

Mots-clés : mot1, mot2, mot3, mot4, mot5.

Summary

Abstract: Quisque ullamcorper placerat ipsum. Cras nibh. Morbi vel justo vitae lacus tincidunt ultrices. Lorem ipsum dolor sit amet, consectetur adipiscing elit. In hac habitasse platea dictumst. Integer tempus convallis augue. Etiam facilisis. Nunc elementum fermentum wisi. Aenean placerat. Ut imperdiet, enim sed gravida sollicitudin, felis odio placerat quam, ac pulvinar elit purus eget enim. Nunc vitae tortor. Proin tempus nibh sit amet nisl. Vivamus quis tortor vitae risus porta vehicula.

Keywords: keyword1, keyword2, keyword3, keyword4, keyword5.

List of Figures

1.1	Oracle MADC Location	3
1.2	MADC Major Milestones	3
2.1	Overview of the Remote Sensing process	10
2.2	Electromagnetic spectrum	11
2.3	Sentinel-2 global coverage	13
2.4	Military Grid Reference System showing the 6° x 8° grid cells	15
2.5	Quadkey hierarchical tile subdivision at increasing zoom levels.	15
2.6	Typical NDVI temporal profile for winter wheat over growing seasons . . .	16
2.7	Crop calendar for Rwanda illustrating Season A, Season B, and Season C with their associated crops and rainfall patterns	17
3.1	Sample figure in Chapter 3	26

List of Tables

2.1	Main Sentinel-2 spectral bands used in agriculture.	14
3.1	Sample table in Chapter 3	26

List of Listings

3.1	Sample Python listing	25
-----	---------------------------------	----

Contents

Dedication	ii
Acknowledgments	iii
Résumé	iv
Summary	v
List of Figures	vi
List of Tables	vii
List of Listings	viii
1 General Context	2
1.1 Introduction	2
1.2 Presentation of the Host Organization	2
1.2.1 Oracle Corporation	2
1.2.2 Oracle Morocco R&D Center	2
1.2.3 Business Area of Activity	4
1.2.3.1 Oracle Morocco R&D Center Lines of Businesses (LOBs)	4
1.2.3.1.1 Oracle APEX	4
1.2.3.1.2 Oracle Health and Artificial Intelligence	4
1.2.3.1.3 Oracle Linux	4
1.2.3.1.4 Oracle NetSuite	4
1.2.3.1.5 Oracle Database Development Tools	5
1.2.3.1.6 Fusion ERP	5
1.2.3.1.7 Oracle Analytics	5
1.2.3.1.8 Oracle Cloud Infrastructure	5
1.2.3.1.9 Oracle Digital Government Group	5
1.2.3.1.10 Oracle Java Platform Group	5
1.2.4 Organizational Structure	5
1.2.5 Organizational Culture	6
1.3 Topic Context	7
1.3.1 General Context of the Project	7

1.3.2	Problematic	7
1.3.3	Solution	8
1.4	Conclusion	9
2	Literature Review	10
2.1	Introduction	10
2.2	Remote Sensing Fundamentals	10
2.2.1	Electromagnetic Spectrum	11
2.2.2	Interaction with Earth's Surface	11
2.2.3	Remote Sensing Applications in Agriculture	12
2.3	Sentinel-2 Satellite Imagery	12
2.3.1	Overview of the Sentinel-2 Mission	13
2.3.2	Spectral Bands and Agricultural Applications	13
2.3.3	Spatial Indexing and Data Organisation	14
2.3.3.1	The Sentinel-2 Tile System and MGRS Grid	14
2.3.3.2	Quadkey Hierarchical Indexing	15
2.4	Crop Phenology	16
2.4.1	Phenological Cycles and Temporal Profiles	16
2.4.2	Crop Calendar in Rwanda	17
2.5	Literature Review	17
2.5.1	Snapshot-Based Approaches	18
2.5.1.1	Traditional Machine Learning and Ensemble Methods	18
2.5.1.2	Deep Learning Methods for Spatial Classification	18
2.5.1.3	Challenges and Limitations	19
2.5.2	Time-Series Based Approaches	19
2.5.2.1	Classical Time-Series Classification	19
2.5.2.2	Deep Learning for Time-Series: LSTM and GRU	19
2.5.2.3	Phenology-Based Temporal Classification	20
2.5.3	Comparative Analysis and Positioning of Our Work	20
2.6	Conclusion	21
3	Methodology and Design	22
3.1	Section One	22
3.1.1	Subsection One	22
3.1.2	Subsection Two	23
3.1.2.1	Subsubsection One	23
3.1.2.2	Subsubsection Two	23
3.1.2.2.1	Paragraph a	23
3.1.2.2.2	Paragraph b	24

3.2	Section Two	24
3.2.1	Subsection One	24
3.2.2	Subsection Two	24
3.2.2.1	Subsubsection One	25
3.2.2.2	Subsubsection Two	25
3.2.2.2.1	Paragraph a	25
3.2.2.2.2	Paragraph b	25
4	Implementation and Results	27
4.1	Section One	27
4.1.1	Subsection One	27
4.1.2	Subsection Two	27
4.1.2.1	Subsubsection One	28
4.1.2.2	Subsubsection Two	28
4.1.2.2.1	Paragraph a	28
4.1.2.2.2	Paragraph b	28
4.2	Section Two	29
4.2.1	Subsection One	29
4.2.2	Subsection Two	29
4.2.2.1	Subsubsection One	29
4.2.2.2	Subsubsection Two	30
4.2.2.2.1	Paragraph a	30
4.2.2.2.2	Paragraph b	30
5	Evaluation and Discussion	31
5.1	Section One	31
5.1.1	Subsection One	31
5.1.2	Subsection Two	31
5.1.2.1	Subsubsection One	32
5.1.2.2	Subsubsection Two	32
5.1.2.2.1	Paragraph a	32
5.1.2.2.2	Paragraph b	32
5.2	Section Two	33
5.2.1	Subsection One	33
5.2.2	Subsection Two	33
5.2.2.1	Subsubsection One	33
5.2.2.2	Subsubsection Two	34
5.2.2.2.1	Paragraph a	34
5.2.2.2.2	Paragraph b	34

General Conclusion and Perspectives	35
A Glossary	36
B Acronyms	37
C Supplementary Material	38
Bibliography	39

IMPORTANT CONFIDENTIALITY WARNING

This report is an Oracle internal document. It is strictly confidential. However, Soukaina Bouhsissin and the examiners, as members of the ENSAM pedagogical team in charge of supervising this project, are authorized to consult it for the sole purpose of evaluating the author. Any use outside of this context is prohibited. If you have obtained a copy of this report without it being explicitly intended for you by the author, please delete it and report back to the author immediately. In addition, the examples described in this report are for illustrative purposes only and do not reflect Oracle's actual business activities.

General Context

1.1 Introduction

In today's world, businesses, technology companies, and the public sector rely on databases to store their data, while utilizing cloud services to manage their IT infrastructure and other essential tools such as ERP, CRM, SCM, EPM, and SPM. These technologies have taken a crucial role, making it inconceivable to function without them.

Oracle, a leading US multinational computer technology corporation, not only employs these tools in its day-to-day operations but also creates and provides them to customers. Moreover, the company places a high value on innovation and ensuring customer satisfaction.

1.2 Presentation of the Host Organization

1.2.1 Oracle Corporation

Oracle Corporation is a tech company with headquarters in Austin, Texas. In 1977, Larry Ellison, Bob Miner, and Ed Oates founded Oracle Corporation, then called Software Development Laboratories (SDL).

The company first focused on creating Oracle Database, a relational database management system (RDBMS), that became its flagship product. Enterprise software, cloud computing, hardware systems, and consulting services are now all part of Oracle's growing list of services and product options. Sun Microsystems was a key acquisition in 2010, giving Oracle access to technologies such as Java and the Solaris operating system.

Oracle is a leading provider of cloud-based IT infrastructure and software that helps organizations grow, find new sources of efficiency, and improve their performance. Oracle developed the first and only autonomous database in the world as well as Oracle Cloud.

1.2.2 Oracle Morocco R&D Center

The first Oracle R&D center in Africa and one of only a few globally, the Oracle Research & Development Center Morocco is a software development center established by Oracle Corporation in Casablanca, Morocco. The center was inaugurated in 2022 and is one of Oracle's key development centers in the EMEA region. The center is staffed by highly skilled



Figure 1.1: Oracle MADC Location

and talented software developers, engineers, and project managers who work on a variety of projects across various Oracle product lines utilizing the newest trends in innovation, such as artificial intelligence, augmented/virtual/mixed reality, big data analytics, Blockchain, cloud computing, cybersecurity, internet of things, machine learning, mobile computing, and serverless computing.

The center's researchers make use of all these technologies to tackle the most pressing problems facing business, science, and the public sector. Expansion of internship and graduate recruitment initiatives and joint research projects with regional universities are all part of this global program.

Oracle MADC has been successful in delivering several high-quality products and solutions, such as Oracle Cloud Infrastructure (OCI), Autonomous Data Warehouse (ADW), etc.

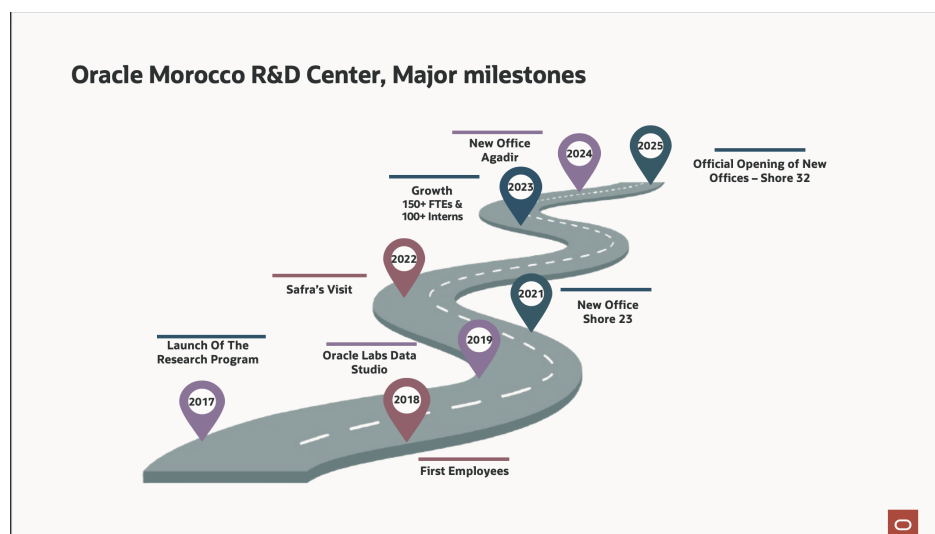


Figure 1.2: MADC Major Milestones

1.2.3 Business Area of Activity

Oracle provides a variety of goods and services to its clients. Software development tools, cloud services, training, consultancy, and credentials are just a few of the goods and services offered. One of the brands Oracle offers is Oracle Database, which has held the top spot since 1979 and is supported by cloud artificial intelligence. SaaS, PaaS, IaaS, and DaaS are just a few of the integrated solutions offered by Oracle Cloud Infrastructure for business, IT, and development needs. Oracle Middleware is a platform integrated for creating and running intelligent, agile applications while increasing technical efficiency using contemporary software designs. Oracle Services include Oracle Advanced Customer Support Services, Oracle Premium Support, Oracle Consulting, Oracle Financing, and Oracle Managed Cloud Services.

1.2.3.1 Oracle Morocco R&D Center Lines of Businesses (LOBs)

1.2.3.1.1 Oracle APEX The most widely used enterprise low-code application platform in the world, Oracle APEX lets you create safe, scalable, and feature-rich web and mobile apps that can be deployed on-premises or in the cloud. With APEX, developers can create and launch engaging applications that address practical issues and deliver benefits right away. It is not necessary to possess extensive knowledge of multiple technologies to provide complex solutions.

1.2.3.1.2 Oracle Health and Artificial Intelligence Oracle APEX helps transform the technologies that are revolutionizing healthcare. Oracle APEX solutions accelerate the development of the next generation of drugs, devices, and therapies through a cohesive ecosystem that harnesses data to deliver results. There are no more silos — just clinical trials, data, strategy, and insights flowing together for better outcomes.

1.2.3.1.3 Oracle Linux Oracle Linux delivers the operating system of choice for Oracle customers, Oracle Cloud, and Oracle Engineered Systems. Oracle has been an active contributor since 1998, shipping the first-ever commercial database on Linux along with multiple enhancements. Today, it is focused on developing open-source products while improving the stability and performance of the operating system.

1.2.3.1.4 Oracle NetSuite NetSuite is a cloud-based business management software that helps organizations manage their financials, operations, and customer relationships. It offers features such as accounting, CRM, inventory management, e-commerce, and more, all in one platform.

1.2.3.1.5 Oracle Database Development Tools The cloud and on-premises solutions drive the secure data access at a scale that powers the modern world. Combined with advanced data visualization and analysis, Oracle Database Development Tools enable data platforms for application development for any purpose. The products include Oracle SQL Developer, Oracle Data Modeler, Oracle REST Data Services, and Oracle SQLcl.

1.2.3.1.6 Fusion ERP Oracle Fusion Cloud ERP is a complete, modern, cloud ERP suite that provides teams with advanced capabilities, such as AI to automate manual processes, analytics to react to market shifts in real-time, and automatic updates to stay current and gain a competitive advantage.

1.2.3.1.7 Oracle Analytics Oracle Analytics is a comprehensive platform for diverse data workloads. It turns information into action with insights from cloud, on-premises, and hybrid data, empowering users to make business-defining decisions sooner through AI and ML-powered analytical applications.

1.2.3.1.8 Oracle Cloud Infrastructure Oracle Cloud is the first public cloud built from the ground up to be a better cloud for every application. By rethinking core engineering and systems design for cloud computing, it created innovations that solve problems customers have with existing public clouds and offers the complete services customers need to build innovative cloud applications.

1.2.3.1.9 Oracle Digital Government Group The data intelligence platform provides the insights that government leaders need to make the decisions that matter. The mission is to help national governments on their digital transformation journeys and improve citizens' experience with government services.

1.2.3.1.10 Oracle Java Platform Group The Oracle Java Platform Group is dedicated to advancing and maintaining the Java platform, a cornerstone of modern software development used by millions of developers worldwide. This group drives innovations in the Java ecosystem, ensuring the language remains robust, efficient, and versatile for a wide range of applications.

1.2.4 Organizational Structure

Oracle Corporation has a hierarchical organizational structure with three main business divisions: Hardware, Services, Cloud, and License. Each segment is divided into smaller business units focused on goods, services, or geographical areas. The Board of Directors oversees the company's performance and strategic direction, while the Executive Leadership

Team manages daily operations. Functional divisions, including sales, marketing, finance, human resources, legal, and IT, fall under the Executive Leadership Team.

Business units are arranged by region, product, and service, with a senior VP or general manager overseeing each unit's operations. The employees are subordinated to both a functional manager and a business unit manager in the matrix-based organizational structure of the business units.

In general, Oracle's organizational structure is built to encourage teamwork, creativity, and client attention. The matrix form enables people to work across many functions and business divisions to achieve shared goals, while the hierarchical structure promotes effective decision-making and communication.

1.2.5 Organizational Culture

Oracle has a strong corporate culture that prioritizes innovation, client focus, and excellence. The company's history of innovation, emphasis on customer satisfaction, and dedication to fostering a friendly and collaborative work environment all shape the company's culture.

Oracle's culture places a strong emphasis on the following values:

Innovation One of Oracle's defining characteristics. The business has a lengthy history of creating cutting-edge goods and technology that have revolutionized the tech sector, promoted by a dedication to R&D and by giving staff members the tools to test fresh concepts.

Customer Satisfaction Oracle's customer-centric approach reflects its commitment to providing goods and services that satisfy client needs. Employees are encouraged to collaborate closely with clients to comprehend their demands and create tailored solutions.

Quality Oracle is dedicated to providing quality operations, products, and services, reflected in its solid reputation for dependability and performance. The company expects every person to strive for excellence.

Collaboration and Teamwork Effective collaboration is essential for achieving Oracle's goals. The company encourages employees to work together across departments and functions, providing tools and resources to facilitate communication.

Overall, Oracle's organizational culture is characterized by a commitment to innovation, customer focus, excellence, collaboration, and teamwork. These values are reflected in the company's products, services, and operations, and are embraced by its employees at all levels of the organization.

1.3 Topic Context

1.3.1 General Context of the Project

Before diving into the core issue of the topic, it is useful to provide some background information. The team I worked with during this internship was called “Digital Government” — which directly reflects the primary objective this project aims to serve.

Digital government services — also referred to as e-government — refer to the use of information and communication technologies to improve the efficiency, effectiveness, and transparency of government operations. These systems reduce bureaucratic delays, minimise paperwork, and enhance the overall quality of public services.

E-government systems are commonly categorised based on the type of interaction they facilitate. Three main categories exist: Government-to-Government (G2G), Government-to-Citizen (G2C), and Government-to-Employee (G2E).

G2G refers to interactions between different government agencies or departments, whether at the same level or across levels of government. An example relevant to this project is the sharing of agricultural satellite data between a national space agency and a ministry of agriculture to inform food security decisions.

G2C focuses on delivering government services and information directly to citizens — making them more accessible, timely, and actionable. In the context of agriculture, this translates to providing smallholder farmers and local administrations with accurate, up-to-date crop production intelligence.

G2E refers to internal government process management and employee communication — for example, equipping field agents with data-driven tools to support their monitoring activities.

The Digital Government team at Oracle aims to leverage existing government data and modern technologies to solve real-world problems across sectors including agriculture, healthcare, and education. This project sits firmly within the agricultural domain and contributes to Oracle’s Government Data Intelligence for Agriculture platform — launched in September 2025 — which supports governments in building more resilient and data-driven food systems.

It is important to note that this project is part of an ongoing collaboration with different governments. The requirements and needs of each country differ, and Rwanda represents one of the early adopters of this approach.

1.3.2 Problematic

During my internship, my focus was on developing a machine learning pipeline to classify crop types across Rwanda using Sentinel-2 satellite time series data — with the broader aim

of supporting agricultural monitoring and food security decision-making.

Agriculture plays a vital role in Rwanda's economy and food system. Accurate, timely knowledge of what crops are growing where — and how much — is essential for effective resource allocation, early warning of food shortfalls, and informed policy decisions. However, obtaining this information at national scale through traditional ground surveys is physically and financially impossible. Surveys are expensive, sparse, seasonal, and always outdated by the time results are available.

The challenge is compounded by the nature of smallholder agriculture in Rwanda. Farms are small, fragmented, and often intercropped — with multiple crops growing on the same plot within the same season. Satellite pixels at the boundary between fields capture mixed spectral signals from more than one crop, creating label ambiguity. Cloud cover — persistent throughout Rwanda's growing seasons — introduces temporal gaps in satellite observations that must be handled carefully. And the extreme class imbalance between dominant crops and minority crops like Sweet Potato — representing only 0.3% of the training data — makes reliable multi-class classification a significant methodological challenge.

1.3.3 Solution

To address these challenges, I designed and implemented an end-to-end multi-class crop classification pipeline using Sentinel-2 satellite time series data, processed on Oracle Cloud Infrastructure.

The pipeline begins with rigorous data preprocessing — including label noise correction, boundary pixel identification and removal, geometric quality filtering, cloud-based quality flagging, temporal gap analysis, and linear interpolation to reconstruct missing observations. Each step was grounded in peer-reviewed literature and validated against the data.

Eight vegetation indices were computed from the cleaned time series — capturing greenness, soil-adjusted vegetation, chlorophyll content, and moisture levels. From these, 50 summary features were extracted per pixel: 32 statistical features capturing the overall seasonal signal, and 18 phenological features capturing crop growth timing — start of season, end of season, peak, amplitude, rate of increase, and integral.

Feature selection was applied in three stages — ANOVA, Random Forest importance ranking, and Recursive Feature Elimination — reducing the feature set from 50 to 37 without loss of predictive performance. Class imbalance was addressed through a combination of balanced class weights, SMOTE oversampling of the minority class, and prediction threshold adjustment — all optimised through grid search.

Two models were trained and compared: Random Forest as a baseline and XGBoost as the primary model. XGBoost outperformed Random Forest across all evaluation metrics — Macro F1, Weighted F1, G-mean, and Cohen's Kappa — with particularly significant gains on Sweet Potato classification, where F1 improved from 0.35 to 0.64.

This approach demonstrated that reliable multi-class crop mapping is achievable from free satellite data at national scale — with no field surveys required — and provides a foundation for integration into Oracle’s Government Data Intelligence for Agriculture platform.

1.4 Conclusion

This chapter introduced the general context of the internship project. It first presented Oracle as the host organization, including its activities, structure, culture, and the role of the Oracle Morocco Research and Development Center. It then described the Digital Government context in which the project was conducted, highlighting the importance of e-government services and their role in improving public-sector decision-making.

The chapter also presented the main problem addressed during the internship: the difficulty of producing accurate and timely crop-type information at national scale using traditional field-survey methods. Finally, it introduced the proposed satellite-based machine learning solution, which aims to support agricultural monitoring and food security decision-making in Rwanda.

Literature Review

2.1 Introduction

This chapter reviews the theoretical foundations and existing literature relevant to crop mapping using satellite imagery and machine learning techniques. It begins by introducing the principles of remote sensing and the characteristics of Sentinel-2 imagery that make it suitable for agricultural applications. It then examines crop phenology as a key concept underlying time-series-based classification.

Finally, it reviews existing approaches to crop mapping from snapshot-based methods to time-series and deep learning techniques identifying their strengths, limitations, and the gaps that motivate the approach adopted in this project.

2.2 Remote Sensing Fundamentals

Remote sensing refers to the acquisition of information about an object, surface, or phenomenon without direct physical contact. In Earth Observation, remote sensing systems mounted on satellites or aircraft measure the electromagnetic radiation reflected or emitted by the Earth's surface in order to characterize natural and human activities.

Remote sensing technologies have become fundamental tools for environmental monitoring, land-cover analysis, climate studies, and agricultural management. By providing repeated and large-scale observations of the Earth's surface, satellite imagery enables continuous monitoring of vegetation dynamics and crop development over time.

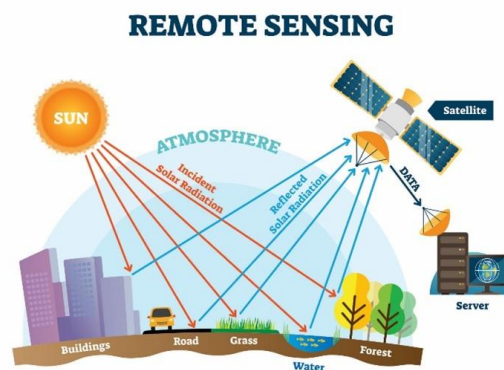


Figure 2.1: Overview of the Remote Sensing process

2.2.1 Electromagnetic Spectrum

Electromagnetic radiation is a form of energy that propagates through space in the form of waves. The electromagnetic spectrum encompasses a wide range of wavelengths, extending from gamma rays and X-rays to visible light, infrared, and radio waves.

Remote sensing applications primarily rely on the **visible**, **near-infrared (NIR)**, and **shortwave infrared (SWIR)** portions of the spectrum, as these bands contain valuable information about vegetation and soil characteristics.

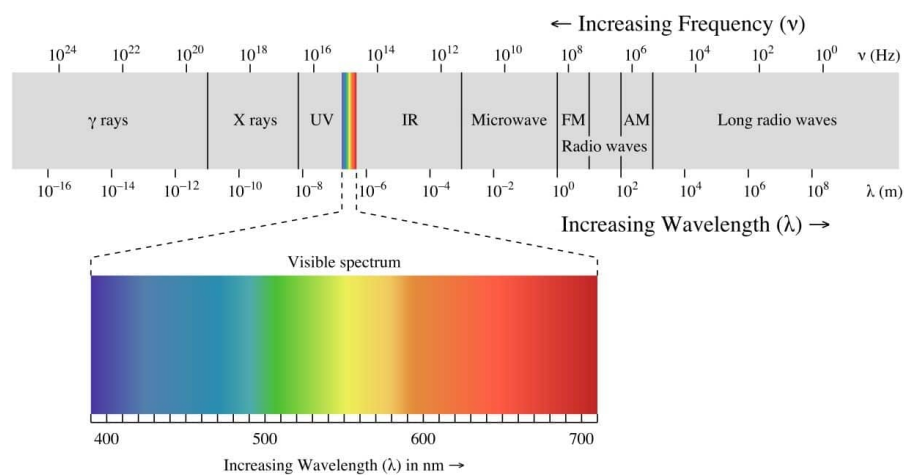


Figure 2.2: Electromagnetic spectrum

Visible light corresponds to wavelengths approximately between 400 nm and 700 nm and includes the blue, green, and red bands perceived by the human eye. Vegetation strongly absorbs red light due to chlorophyll activity while reflecting green light, which explains the green appearance of plants.

Near-infrared radiation, typically between 700 nm and 1300 nm, is highly reflected by healthy vegetation because of the internal cellular structure of plant leaves. This property makes the NIR region particularly important for vegetation monitoring and crop analysis.

Shortwave infrared bands are sensitive to vegetation moisture content, soil properties, and water stress conditions. These bands are frequently used in agricultural and environmental monitoring applications.

The spectral response of different land-cover types varies across wavelengths, allowing satellite sensors to distinguish vegetation, water bodies, urban areas, and bare soil based on their spectral signatures.

2.2.2 Interaction with Earth's Surface

When electromagnetic radiation reaches the Earth's surface, it is either **absorbed**, **transmitted**, or **reflected**. The proportion of each depends on the wavelength and the physical and chemical

properties of the surface material. This wavelength-dependent reflectance behaviour is referred to as a **spectral signature**, and it forms the physical basis for distinguishing different land cover types from satellite imagery [1].

Vegetation exhibits a distinctive spectral signature: strong absorption in the red band (around 670 nm) driven by chlorophyll activity, and high reflectance in the near-infrared (NIR) region (around 800 nm) due to the internal structure of leaf cells. Bare soil, water, and built surfaces each exhibit markedly different signatures, enabling automated classification algorithms to discriminate between them. Within vegetation itself, spectral differences between crop types — though subtler — can be exploited when sufficient spectral and temporal resolution is available [2].

2.2.3 Remote Sensing Applications in Agriculture

Remote sensing has become an essential technology for modern **agricultural monitoring** and **precision farming**. Satellite imagery enables large-scale and frequent observation of agricultural areas without the need for costly field surveys.

In agriculture, remote sensing is commonly used for **crop type mapping**, **vegetation health assessment**, **yield estimation**, **drought monitoring**, **irrigation management**, and detection of environmental **stress conditions**.

The temporal dimension provided by satellite time series is particularly important for crop classification tasks. Different crops exhibit distinct seasonal growth patterns, which can be observed through changes in spectral reflectance over time. Monitoring these temporal dynamics enables more accurate identification of crop types compared to single-date observations.

Despite its advantages, agricultural remote sensing also faces several challenges. Persistent cloud cover may reduce image availability, especially in tropical regions. Mixed pixels caused by fragmented agricultural parcels can introduce spectral ambiguity, while variations in planting dates and farming practices increase the complexity of crop classification tasks.

2.3 Sentinel-2 Satellite Imagery

Sentinel-2 is one of the most important Earth Observation missions developed within the **European Union's Copernicus Programme**, specifically designed for land monitoring applications. Its high temporal resolution, rich spectral information, and free open-access policy make it a de facto standard for large-scale crop classification tasks based on satellite time series.

2.3.1 Overview of the Sentinel-2 Mission

The Sentinel-2 mission consists of two polar-orbiting satellites: **Sentinel-2A**, launched in 2015, and **Sentinel-2B**, launched in 2017. Both satellites operate in the same sun-synchronous orbit at an altitude of approximately 786 km. The combination of the two satellites enables a revisit frequency of approximately **five days** at the equator and even shorter revisit times at higher latitudes. This frequent acquisition capability is essential for monitoring dynamic agricultural processes and capturing crop evolution throughout the growing season [3].

Sentinel-2 provides **multispectral imagery** in thirteen spectral bands covering the visible, near-infrared (NIR), and shortwave infrared (SWIR) regions of the electromagnetic spectrum, at spatial resolutions of **10 m, 20 m, and 60 m** depending on the spectral band. Its free and open-access policy has significantly accelerated research and operational applications in remote sensing and agricultural monitoring worldwide.

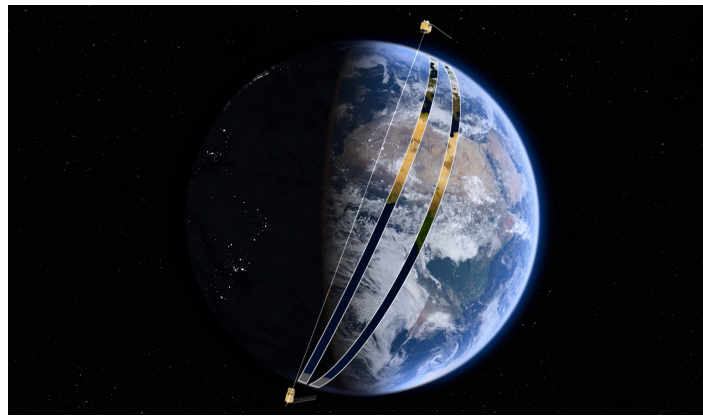


Figure 2.3: Sentinel-2 global coverage

2.3.2 Spectral Bands and Agricultural Applications

The Sentinel-2 sensor acquires data in **thirteen spectral** bands specifically designed for vegetation, soil, and water monitoring applications. These bands capture different physical and biochemical properties of vegetation and are widely used for crop analysis.

The **visible bands** (Blue, Green, and Red) are mainly used for visual interpretation and vegetation analysis. The **near-infrared band** is highly sensitive to plant vigor and biomass, while the **red-edge** bands provide valuable information about chlorophyll concentration and vegetation stress. The **shortwave infrared** bands are useful for monitoring moisture content and soil conditions.

Table 2.1 summarizes the main Sentinel-2 spectral bands commonly used in agricultural applications.

Table 2.1: Main Sentinel-2 spectral bands used in agriculture.

Band	Description	Wavelength (nm)	Resolution
B2	Blue	490	10 m
B3	Green	560	10 m
B4	Red	665	10 m
B5	Red Edge 1	705	20 m
B6	Red Edge 2	740	20 m
B7	Red Edge 3	783	20 m
B8	Near Infrared (NIR)	842	10 m
B8A	Narrow NIR	865	20 m
B11	SWIR 1	1610	20 m
B12	SWIR 2	2190	20 m

The availability of red-edge bands represents one of the key advantages of Sentinel-2 for agricultural monitoring. These bands improve the detection of vegetation stress and enhance crop discrimination capabilities compared to older satellite missions.

2.3.3 Spatial Indexing and Data Organisation

Efficient spatial organisation of satellite imagery is essential for large-scale time-series analysis. Two complementary indexing systems are relevant to this project: the MGRS tile grid used natively by Sentinel-2, and the quadkey hierarchical indexing scheme used for cloud-based data management.

2.3.3.1 The Sentinel-2 Tile System and MGRS Grid

Sentinel-2 imagery is organised and distributed using the **Military Grid Reference System** (MGRS), a global spatial indexing framework built on top of the **Universal Transverse Mercator** (UTM) projection [4]. The UTM system divides the Earth into 60 longitudinal zones of 6 degrees each, numbered 1 to 60 from west to east. Within each zone, coordinates are expressed in metres — easting and northing — relative to a zone-specific central meridian, minimising distortion within each zone and ensuring geometric consistency for pixel-level spatial operations.

Under the MGRS system, each UTM zone is further subdivided into a grid of 100×100 km tiles, each identified by a unique alphanumeric code composed of the UTM zone number, a latitude band letter, and a two-letter square identifier — for example, 36MTC and 36MTD for tiles covering Rwanda [4]. Each Sentinel-2 acquisition is projected and delivered within this fixed tile grid, ensuring that imagery from different dates is inherently co-registered and spatially aligned — a prerequisite for constructing pixel-wise time series.

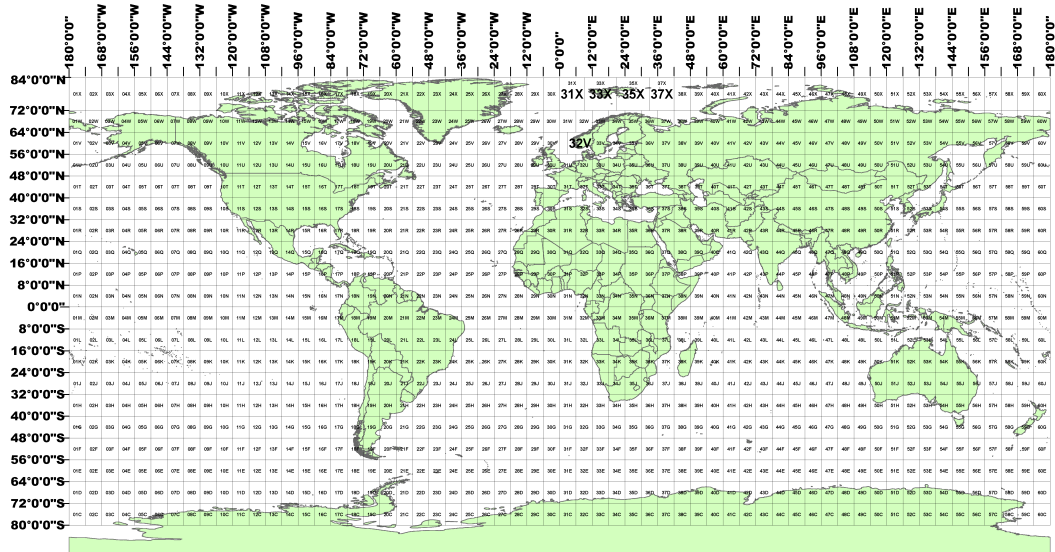


Figure 2.4: Military Grid Reference System showing the 6° x 8° grid cells

2.3.3.2 Quadkey Hierarchical Indexing

Beyond the MGRS tile system, modern large-scale geospatial platforms increasingly rely on hierarchical spatial indexing schemes to efficiently store, query, and retrieve satellite data at scale. One of the most widely adopted schemes is the **quadkey system**, originally introduced by Microsoft for web mapping applications [5].

The quadkey system subdivides the Earth's surface recursively into a quadtree of tiles at increasing zoom levels. At each level, every tile is divided into four equal quadrants — top-left (0), top-right (1), bottom-left (2), and bottom-right (3). A tile at any zoom level is uniquely identified by a string of digits encoding its hierarchical position. For example, a tile at zoom level 3 might carry the quadkey 021, meaning it occupies the second quadrant of the first quadrant of the third quadrant of the root tile.

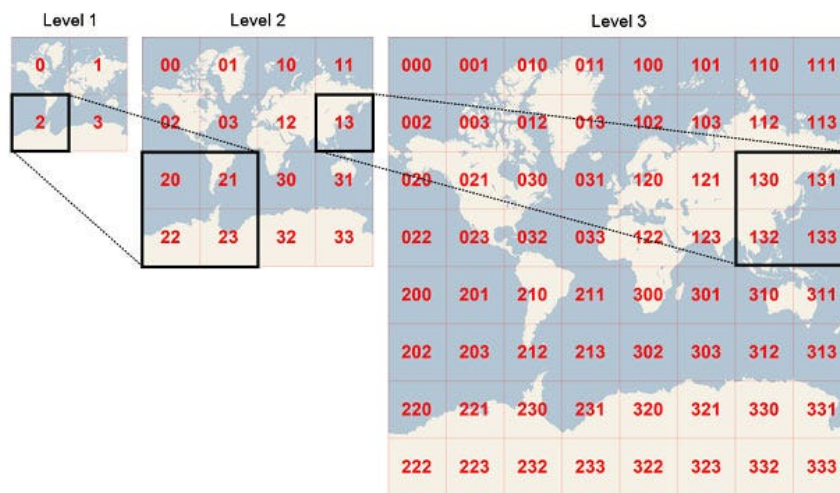


Figure 2.5: Quadkey hierarchical tile subdivision at increasing zoom levels.

This recursive structure offers several practical advantages for large-scale geospatial data management. The multi-resolution hierarchy also naturally supports data aggregation across spatial scales [5]. In the context of this project, quadkey indexing is used to partition Sentinel-2 time series data enabling efficient tile-based processing and scalable retrieval of pixel observations across Rwanda’s territory.

2.4 Crop Phenology

Crop phenology refers to the study of cyclic and seasonal biological events in crops and their timing in relation to climate and environmental conditions [6]. From a remote sensing perspective, phenology is the key signal that allows satellite time series to discriminate between crop types that may appear spectrally similar at any single point in time but follow distinctly different growth trajectories over a full growing season.

2.4.1 Phenological Cycles and Temporal Profiles

Each crop type follows a characteristic growth cycle — from germination and early vegetative growth, through canopy closure, flowering, and grain fill, to senescence and harvest. This cycle produces a distinctive temporal profile of spectral reflectance when observed repeatedly from satellite, commonly referred to as a phenological curve [7].

These temporal profiles are the fundamental signal exploited by time-series classification approaches. Crops that are spectrally similar at a single date may exhibit clearly distinct seasonal trajectories — differing in the timing of green-up, peak greenness, amplitude, and senescence date — making temporal information essential for accurate multi-class crop mapping [8]. Key phenological metrics such as start of season, end of season, peak greenness, seasonal amplitude, and seasonal integral serve as compact and discriminative features encoding crop-type-specific growth dynamics.

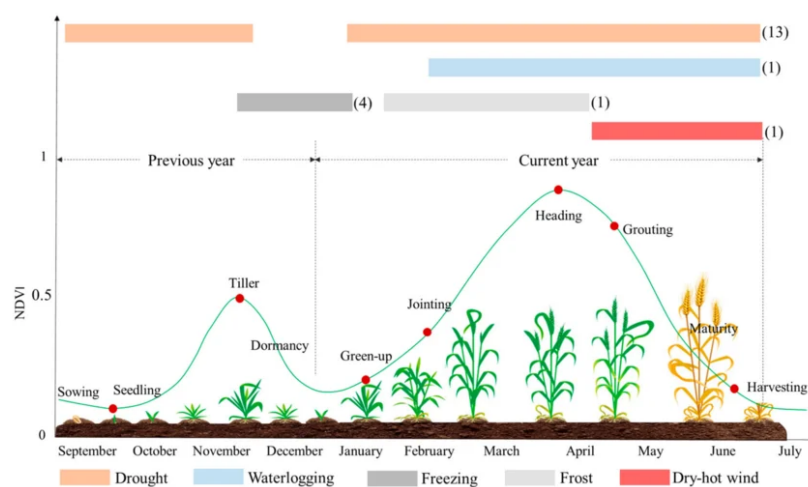


Figure 2.6: Typical NDVI temporal profile for winter wheat over growing seasons

2.4.2 Crop Calendar in Rwanda

Agriculture in Rwanda is primarily organised around two major cropping seasons, commonly referred to as **Season A** and **Season B**, strongly influenced by rainfall patterns and climatic conditions [9]. Season A generally begins in September and ends in February, while Season B extends from February to August. Certain regions also support an additional short agricultural season known as Season C, typically practised in irrigated areas between July and August.

The temporal distribution of planting and harvesting periods varies across crop types and geographical regions, reflecting Rwanda's diverse agro-ecological zones. Major crops cultivated in Rwanda include **maize, beans, cassava, Irish potato, rice, banana, and sweet potato** [9]. Understanding the crop calendar is essential for interpreting satellite time-series data, as crop spectral responses are directly linked to their phenological development stages. The overlap between rainy seasons and cropping periods also introduces persistent cloud cover, creating temporal gaps in Sentinel-2 observations that must be addressed in the preprocessing pipeline.

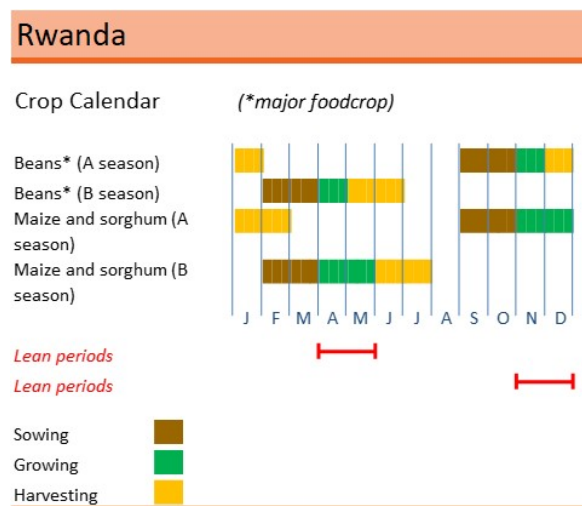


Figure 2.7: Crop calendar for Rwanda illustrating Season A, Season B, and Season C with their associated crops and rainfall patterns

2.5 Literature Review

Crop type mapping using satellite imagery has been an active area of research over the past two decades, driven by advances in satellite data availability, machine learning, and cloud computing. This section reviews the existing literature relevant to this project, organised around three major themes: snapshot-based classification approaches, time-series-based methods, and the specific challenges of rare class detection and imbalanced learning in agricultural remote sensing. For each theme, key contributions are synthesised, their limitations identified, and their relevance to the approach adopted in this project discussed.

2.5.1 Snapshot-Based Approaches

Early work in satellite-based crop mapping relied predominantly on single-date or snapshot imagery, where classification is performed using spectral information from one or a small number of acquisition dates. While computationally straightforward, these approaches are fundamentally limited by the fact that a single image captures only one moment in a crop's growth cycle, making it difficult to distinguish spectrally similar crops that differ primarily in their temporal behaviour.

2.5.1.1 Traditional Machine Learning and Ensemble Methods

Traditional machine learning methods — particularly **Support Vector Machines** (SVM), **Random Forest** (RF), and **gradient boosting** approaches — have been widely applied to crop mapping using single-date or multi-date optical imagery [10]. These methods operate on handcrafted spectral features derived from individual bands or vegetation indices, and have demonstrated competitive performance in regions with simple cropping systems and low cloud cover.

Random Forest in particular has become a standard baseline for crop classification tasks due to its robustness to noisy features, ability to handle high-dimensional inputs, and built-in feature importance estimation [11]. **Gradient boosting** methods such as **XGBoost** have more recently demonstrated superior performance in tabular classification tasks, achieving state-of-the-art results on several remote sensing benchmarks [12]. However, both approaches applied to single-date imagery are inherently limited: they cannot capture the temporal dynamics that distinguish crop types with similar spectral signatures at any given moment.

2.5.1.2 Deep Learning Methods for Spatial Classification

The advent of deep learning brought convolutional neural networks (CNNs) to remote sensing, enabling the automatic extraction of spatial features from satellite imagery without manual feature engineering [13]. CNN-based approaches applied to single-date imagery have shown strong performance in land cover mapping tasks, particularly when large labelled datasets are available [14].

For crop mapping specifically, CNNs have been applied to high-resolution imagery to exploit spatial context — field shape, texture, and neighbourhood relationships — as complementary discriminative information beyond spectral signatures alone [15]. Despite these advances, CNN-based snapshot approaches share the same fundamental limitation as traditional methods: they remain blind to the temporal dimension of crop growth, which is often the most discriminative signal for crop type identification.

2.5.1.3 Challenges and Limitations

Snapshot-based approaches face several inherent limitations that restrict their applicability in complex agricultural contexts such as Rwanda. First, the reliance on a single acquisition date introduces high sensitivity to atmospheric conditions, cloud cover, and phenological variability between years — a single cloudy image can render an entire classification pipeline unusable [7]. Second, spectrally similar crops that differ primarily in their growth timing cannot be reliably distinguished from a single observation. Third, the class imbalance typical of smallholder landscapes — where dominant crops vastly outnumber minority crops — is particularly difficult to address without the additional discriminative power provided by temporal information. These limitations motivated a shift toward time-series-based approaches.

2.5.2 Time-Series Based Approaches

The availability of dense satellite time series — made possible by missions such as Sentinel-2 — has enabled a fundamental shift in crop mapping methodology. Rather than relying on spectral snapshots, time-series approaches exploit the full temporal trajectory of crop growth, capturing phenological dynamics that are highly discriminative across crop types [7].

2.5.2.1 Classical Time-Series Classification

Early time-series classification methods for crop mapping were based on template matching and distance-based approaches. **Dynamic Time Warping** (DTW), which measures similarity between temporal sequences by allowing elastic alignment in time, was among the first methods applied to satellite phenological curves for crop discrimination [16]. While effective for simple cropping systems, DTW-based methods scale poorly to large datasets and struggle with noisy or gap-filled time series.

Threshold-based phenological methods — which classify pixels based on the timing and magnitude of key phenological events such as start of season, peak, and end of season — have also been widely used for crop mapping in data-sparse environments [17]. These approaches are interpretable and computationally efficient but require careful calibration and are sensitive to temporal gaps introduced by cloud cover.

2.5.2.2 Deep Learning for Time-Series: LSTM and GRU

Recurrent neural networks (RNNs), and in particular **Long Short-Term Memory** (LSTM) networks and **Gated Recurrent Units** (GRUs), have emerged as the dominant deep learning architectures for satellite time-series classification [18]. Unlike classical methods, these architectures learn temporal dependencies directly from data, without requiring explicit phenological feature engineering.

Rußwurm and Körner [19] were among the first to apply LSTM networks to Sentinel-2 time series for crop mapping, demonstrating significant improvements over Random Forest baselines on European agricultural landscapes. Subsequent studies confirmed that recurrent architectures consistently outperform classical machine learning methods when sufficient temporal observations are available, particularly for minority crop classes with distinctive but short-duration phenological signals [20].

GRU-based models have been shown to achieve performance comparable to LSTM while requiring fewer parameters and training time, making them particularly attractive for large-scale operational applications [21]. More recently, attention-based mechanisms and transformer architectures have been explored for satellite time-series classification, though their advantage over well-tuned recurrent models remains dataset-dependent [22].

2.5.2.3 Phenology-Based Temporal Classification

A complementary line of research has focused on extracting explicit phenological features from satellite time series — such as start of season, end of season, peak greenness, seasonal amplitude, and seasonal integral — and using these as inputs to classical machine learning classifiers [7]. This hybrid approach combines the interpretability of phenological metrics with the discriminative power of supervised learning, and has demonstrated strong performance in smallholder contexts where training data is limited [8].

Studies applying this approach to sub-Saharan African agricultural landscapes have highlighted the importance of carefully handling temporal gaps introduced by cloud cover before extracting phenological features, as gap-filling quality directly affects the reliability of derived metrics [23].

2.5.3 Comparative Analysis and Positioning of Our Work

The reviewed literature highlights a clear evolution in crop type mapping methodology — from simple snapshot-based approaches relying on single-date spectral information, to sophisticated time-series methods that exploit the full temporal trajectory of crop growth. While significant progress has been made, several challenges remain insufficiently addressed, particularly in the context of smallholder agriculture in sub-Saharan Africa.

Most existing studies are conducted in temperate agricultural landscapes characterised by large, regular fields and relatively low cloud cover. Few works explicitly address the combination of challenges present in Rwanda: a bimodal cropping calendar, persistent cloud cover during both growing seasons, extreme field fragmentation, and severe class imbalance between dominant and minority crops.

Furthermore, the majority of published pipelines report overall accuracy metrics that are dominated by majority classes, masking poor performance on rare crop types. The challenge

of detecting minority crops — which are often the most important from a food security perspective — remains an open problem that receives limited attention in the literature.

This project is positioned at the intersection of these gaps. It targets a real-world, national-scale crop mapping problem in Rwanda, explicitly addressing the challenges of cloud cover, class imbalance, and rare crop detection within a single end-to-end pipeline. By comparing both classical machine learning and deep learning approaches on the same dataset and evaluation framework, it contributes a rigorous and reproducible benchmark for crop mapping in challenging smallholder contexts.

2.6 Conclusion

This chapter presented the theoretical foundations and existing literature relevant to crop type mapping using Sentinel-2 satellite time series. The first part introduced the principles of remote sensing, the characteristics of Sentinel-2 imagery, and the role of crop phenology as the key temporal signal for crop discrimination. the second part reviewed existing approaches to crop mapping, from snapshot-based methods to time-series and deep learning techniques, identifying their strengths and limitations.

The review identified three key gaps in the existing literature: insufficient attention to rare class detection, the widespread neglect of spatial leakage through boundary pixels, and a lack of studies addressing Rwanda’s specific agricultural context. The methodology presented in the following chapter is designed to address each of these gaps through a rigorous, end-to-end crop classification pipeline built on Sentinel-2 time series data processed on Oracle Cloud Infrastructure.

Methodology and Design

Class aptent taciti sociosqu ad litora torquent per conubia nostra, per inceptos hymenaeos. Aenean nonummy turpis id odio. Integer euismod imperdiet turpis. Ut nec leo nec diam imperdiet lacinia. Etiam eget lacus eget mi ultricies posuere. In placerat tristique tortor. Sed porta vestibulum metus. Nulla iaculis sollicitudin pede. Fusce luctus tellus in dolor. Curabitur auctor velit a sem. Morbi sapien. Class aptent taciti sociosqu ad litora torquent per conubia nostra, per inceptos hymenaeos. Donec adipiscing urna vehicula nunc. Sed ornare leo in leo. In rhoncus leo ut dui. Aenean dolor quam, volutpat nec, fringilla id, consectetur vel, pede.

3.1 Section One

Nulla malesuada risus ut urna. Aenean pretium velit sit amet metus. Duis iaculis. In hac habitasse platea dictumst. Nullam molestie turpis eget nisl. Duis a massa id pede dapibus ultricies. Sed eu leo. In at mauris sit amet tortor bibendum varius. Phasellus justo risus, posuere in, sagittis ac, varius vel, tortor. Quisque id enim. Phasellus consequat, libero pretium nonummy fringilla, tortor lacus vestibulum nunc, ut rhoncus ligula neque id justo. Nullam accumsan euismod nunc. Proin vitae ipsum ac metus dictum tempus. Nam ut wisi. Quisque tortor felis, interdum ac, sodales a, semper a, sem. Curabitur in velit sit amet dui tristique sodales. Vivamus mauris pede, lacinia eget, pellentesque quis, scelerisque eu, est. Aliquam risus. Quisque bibendum pede eu dolor.

3.1.1 Subsection One

Donec tempus neque vitae est. Aenean egestas odio sed risus ullamcorper ullamcorper. Sed in nulla a tortor tincidunt egestas. Nam sapien tortor, elementum sit amet, aliquam in, porttitor faucibus, enim. Nullam congue suscipit nibh. Quisque convallis. Praesent arcu nibh, vehicula eget, accumsan eu, tincidunt a, nibh. Suspendisse vulputate, tortor quis adipiscing viverra, lacus nibh dignissim tellus, eu suscipit risus ante fringilla diam. Quisque a libero vel pede imperdiet aliquet. Pellentesque nunc nibh, eleifend a, consequat consequat, hendrerit nec, diam. Sed urna. Maecenas laoreet eleifend neque. Vivamus purus odio, eleifend non, iaculis a, ultrices sit amet, urna. Mauris faucibus odio vitae risus. In nisl. Praesent purus. Integer iaculis, sem eu egestas lacinia, lacus pede scelerisque augue, in ullamcorper dolor eros ac lacus. Nunc in libero.

3.1.2 Subsection Two

Fusce suscipit cursus sem. Vivamus risus mi, egestas ac, imperdiet varius, faucibus quis, leo. Aenean tincidunt. Donec suscipit. Cras id justo quis nibh scelerisque dignissim. Aliquam sagittis elementum dolor. Aenean consectetur justo in pede. Curabitur ullamcorper ligula nec orci. Aliquam purus turpis, aliquam id, ornare vitae, porttitor non, wisi. Maecenas luctus porta lorem. Donec vitae ligula eu ante pretium varius. Proin tortor metus, convallis et, hendrerit non, scelerisque in, urna. Cras quis libero eu ligula bibendum tempor. Vivamus tellus quam, malesuada eu, tempus sed, tempor sed, velit. Donec lacinia auctor libero.

3.1.2.1 Subsubsection One

Praesent sed neque id pede mollis rutrum. Vestibulum iaculis risus. Pellentesque lacus. Ut quis nunc sed odio malesuada egestas. Duis a magna sit amet ligula tristique pretium. Ut pharetra. Vestibulum imperdiet magna nec wisi. Mauris convallis. Sed accumsan sollicitudin massa. Sed id enim. Nunc pede enim, lacinia ut, pulvinar quis, suscipit semper, elit. Cras accumsan erat vitae enim. Cras sollicitudin. Vestibulum rutrum blandit massa.

3.1.2.2 Subsubsection Two

Sed gravida lectus ut purus. Morbi laoreet magna. Pellentesque eu wisi. Proin turpis. Integer sollicitudin augue nec dui. Fusce lectus. Vivamus faucibus nulla nec lacus. Integer diam. Pellentesque sodales, enim feugiat cursus volutpat, sem mauris dignissim mauris, quis consequat sem est fermentum ligula. Nullam justo lectus, condimentum sit amet, posuere a, fringilla mollis, felis. Morbi nulla nibh, pellentesque at, nonummy eu, sollicitudin nec, ipsum. Cras neque. Nunc augue. Nullam vitae quam id quam pulvinar blandit. Nunc sit amet orci. Aliquam erat elit, pharetra nec, aliquet a, gravida in, mi. Quisque urna enim, viverra quis, suscipit quis, tincidunt ut, sapien. Cras placerat consequat sem. Curabitur ac diam. Curabitur diam tortor, mollis et, viverra ac, tempus vel, metus.

3.1.2.2.1 Paragraph a Curabitur ac lorem. Vivamus non justo in dui mattis posuere. Etiam accumsan ligula id pede. Maecenas tincidunt diam nec velit. Praesent convallis sapien ac est. Aliquam ullamcorper euismod nulla. Integer mollis enim vel tortor. Nulla sodales placerat nunc. Sed tempus rutrum wisi. Duis accumsan gravida purus. Nunc nunc. Etiam facilisis dui eu sem. Vestibulum semper. Praesent eu eros. Vestibulum tellus nisl, dapibus id, vestibulum sit amet, placerat ac, mauris. Maecenas et elit ut erat placerat dictum. Nam feugiat, turpis et sodales volutpat, wisi quam rhoncus neque, vitae aliquam ipsum sapien vel enim. Maecenas suscipit cursus mi.

3.1.2.2.2 Paragraph b Quisque consectetur. In suscipit mauris a dolor pellentesque consectetur. Mauris convallis neque non erat. In lacinia. Pellentesque leo eros, sagittis quis, fermentum quis, tincidunt ut, sapien. Maecenas sem. Curabitur eros odio, interdum eu, feugiat eu, porta ac, nisl. Curabitur nunc. Etiam fermentum convallis velit. Pellentesque laoreet lacus. Quisque sed elit. Nam quis tellus. Aliquam tellus arcu, adipiscing non, tincidunt eleifend, adipiscing quis, augue. Vivamus elementum placerat enim. Suspendisse ut tortor. Integer faucibus adipiscing felis. Aenean consectetur mattis lectus. Morbi malesuada faucibus dolor. Nam lacus. Etiam arcu libero, malesuada vitae, aliquam vitae, blandit tristique, nisl.

3.2 Section Two

Maecenas accumsan dapibus sapien. Duis pretium iaculis arcu. Curabitur ut lacus. Aliquam vulputate. Suspendisse ut purus sed sem tempor rhoncus. Ut quam dui, fringilla at, dictum eget, ultricies quis, quam. Etiam sem est, pharetra non, vulputate in, pretium at, ipsum. Nunc semper sagittis orci. Sed scelerisque suscipit diam. Ut volutpat, dolor at ullamcorper tristique, eros purus mollis quam, sit amet ornare ante nunc et enim.

3.2.1 Subsection One

Phasellus fringilla, metus id feugiat consectetur, lacus wisi ultrices tellus, quis lobortis nibh lorem quis tortor. Donec egestas ornare nulla. Mauris mi tellus, porta faucibus, dictum vel, nonummy in, est. Aliquam erat volutpat. In tellus magna, porttitor lacinia, molestie vitae, pellentesque eu, justo. Class aptent taciti sociosqu ad litora torquent per conubia nostra, per inceptos hymenaeos. Sed orci nibh, scelerisque sit amet, suscipit sed, placerat vel, diam. Vestibulum nonummy vulputate orci. Donec et velit ac arcu interdum semper. Morbi pede orci, cursus ac, elementum non, vehicula ut, lacus. Cras volutpat. Nam vel wisi quis libero venenatis placerat. Aenean sed odio. Quisque posuere purus ac orci. Vivamus odio. Vivamus varius, nulla sit amet semper viverra, odio mauris consequat lacus, at vestibulum neque arcu eu tortor. Donec iaculis tincidunt tellus. Aliquam erat volutpat. Curabitur magna lorem, dignissim volutpat, viverra et, adipiscing nec, dolor. Praesent lacus mauris, dapibus vitae, sollicitudin sit amet, nonummy eget, ligula.

3.2.2 Subsection Two

Cras egestas ipsum a nisl. Vivamus varius dolor ut dolor. Fusce vel enim. Pellentesque accumsan ligula et eros. Cras id lacus non tortor facilisis facilisis. Etiam nisl elit, cursus sed, fringilla in, congue nec, urna. Cum sociis natoque penatibus et magnis dis parturient montes, nascetur ridiculus mus. Integer at turpis. Cum sociis natoque penatibus et magnis dis parturient montes, nascetur ridiculus mus. Duis fringilla, ligula sed porta fringilla, ligula

wisi commodo felis, ut adipiscing felis dui in enim. Suspendisse malesuada ultrices ante. Pellentesque scelerisque augue sit amet urna. Nulla volutpat aliquet tortor. Cras aliquam, tellus at aliquet pellentesque, justo sapien commodo leo, id rhoncus sapien quam at erat. Nulla commodo, wisi eget sollicitudin pretium, orci orci aliquam orci, ut cursus turpis justo et lacus. Nulla vel tortor. Quisque erat elit, viverra sit amet, sagittis eget, porta sit amet, lacus.

3.2.2.1 Subsubsection One

In hac habitasse platea dictumst. Proin at est. Curabitur tempus vulputate elit. Pellentesque sem. Praesent eu sapien. Duis elit magna, aliquet at, tempus sed, vehicula non, enim. Morbi viverra arcu nec purus. Vivamus fringilla, enim et commodo malesuada, tortor metus elementum ligula, nec aliquet est sapien ut lectus. Aliquam mi. Ut nec elit. Fusce euismod luctus tellus. Curabitur scelerisque. Nullam purus. Nam ultricies accumsan magna. Morbi pulvinar lorem sit amet ipsum. Donec ut justo vitae nibh mollis congue. Fusce quis diam. Praesent tempus eros ut quam.

3.2.2.2 Subsubsection Two

Donec in nisl. Fusce vitae est. Vivamus ante ante, mattis laoreet, posuere eget, congue vel, nunc. Fusce sem. Nam vel orci eu eros viverra luctus. Pellentesque sit amet augue. Nunc sit amet ipsum et lacus varius nonummy. Integer rutrum sem eget wisi. Aenean eu sapien. Quisque ornare dignissim mi. Duis a urna vel risus pharetra imperdiet. Suspendisse potenti.

3.2.2.2.1 Paragraph a Morbi justo. Aenean nec dolor. In hac habitasse platea dictumst. Proin nonummy porttitor velit. Sed sit amet leo nec metus rhoncus varius. Cras ante. Vestibulum commodo sem tincidunt massa. Nam justo. Aenean luctus, felis et condimentum lacinia, lectus enim pulvinar purus, non porta velit nisl sed eros. Suspendisse consequat. Mauris a dui et tortor mattis pretium. Sed nulla metus, volutpat id, aliquam eget, ullamcorper ut, ipsum. Morbi eu nunc. Praesent pretium. Duis aliquam pulvinar ligula. Ut blandit egestas justo. Quisque posuere metus viverra pede.

3.2.2.2.2 Paragraph b Vivamus sodales elementum neque. Vivamus dignissim accumsan neque. Sed at enim. Vestibulum nonummy interdum purus. Mauris ornare velit id nibh pretium ultricies. Fusce tempor pellentesque odio. Vivamus augue purus, laoreet in, scelerisque vel, commodo id, wisi. Duis enim. Nulla interdum, nunc eu semper eleifend, enim dolor pretium elit, ut commodo ligula nisl a est. Vivamus ante. Nulla leo massa, posuere nec, volutpat vitae, rhoncus eu, magna.

```
1 # Placeholder code
2 def hello_world():
```

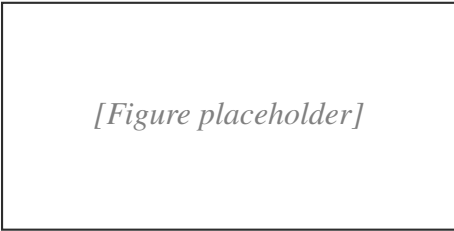


Figure 3.1: Sample figure in Chapter 3

Table 3.1: Sample table in Chapter 3

Column A	Column B	Column C
Item 1	Description	42
Item 2	Description	87

```
3      """Print a greeting."""
4      print("Hello, World!")
5
6 hello_world()
```

Listing 3.1: Sample Python listing

Implementation and Results

Quisque facilisis auctor sapien. Pellentesque gravida hendrerit lectus. Mauris rutrum sodales sapien. Fusce hendrerit sem vel lorem. Integer pellentesque massa vel augue. Integer elit tortor, feugiat quis, sagittis et, ornare non, lacus. Vestibulum posuere pellentesque eros. Quisque venenatis ipsum dictum nulla. Aliquam quis quam non metus eleifend interdum. Nam eget sapien ac mauris malesuada adipiscing. Etiam eleifend neque sed quam. Nulla facilisi. Proin a ligula. Sed id dui eu nibh egestas tincidunt. Suspendisse arcu.

4.1 Section One

Maecenas dui. Aliquam volutpat auctor lorem. Cras placerat est vitae lectus. Curabitur massa lectus, rutrum euismod, dignissim ut, dapibus a, odio. Ut eros erat, vulputate ut, interdum non, porta eu, erat. Cras fermentum, felis in porta congue, velit leo facilisis odio, vitae consectetur lorem quam vitae orci. Sed ultrices, pede eu placerat auctor, ante ligula rutrum tellus, vel posuere nibh lacus nec nibh. Maecenas laoreet dolor at enim. Donec molestie dolor nec metus. Vestibulum libero. Sed quis erat. Sed tristique. Duis pede leo, fermentum quis, consectetur eget, vulputate sit amet, erat.

4.1.1 Subsection One

Donec vitae velit. Suspendisse porta fermentum mauris. Ut vel nunc non mauris pharetra varius. Duis consequat libero quis urna. Maecenas at ante. Vivamus varius, wisi sed egestas tristique, odio wisi luctus nulla, lobortis dictum dolor ligula in lacus. Vivamus aliquam, urna sed interdum porttitor, metus orci interdum odio, sit amet euismod lectus felis et leo. Praesent ac wisi. Nam suscipit vestibulum sem. Praesent eu ipsum vitae pede cursus venenatis. Duis sed odio. Vestibulum eleifend. Nulla ut massa. Proin rutrum mattis sapien. Curabitur dictum gravida ante.

4.1.2 Subsection Two

Phasellus placerat vulputate quam. Maecenas at tellus. Pellentesque neque diam, dignissim ac, venenatis vitae, consequat ut, lacus. Nam nibh. Vestibulum fringilla arcu mollis arcu. Sed et turpis. Donec sem tellus, volutpat et, varius eu, commodo sed, lectus. Lorem ipsum dolor

sit amet, consectetur adipiscing elit. Quisque enim arcu, suscipit nec, tempus at, imperdiet vel, metus. Morbi volutpat purus at erat. Donec dignissim, sem id semper tempus, nibh massa eleifend turpis, sed pellentesque wisi purus sed libero. Nullam lobortis tortor vel risus. Pellentesque consequat nulla eu tellus. Donec velit. Aliquam fermentum, wisi ac rhoncus iaculis, tellus nunc malesuada orci, quis volutpat dui magna id mi. Nunc vel ante. Duis vitae lacus. Cras nec ipsum.

4.1.2.1 Subsubsection One

Morbi nunc. Aliquam consectetur varius nulla. Phasellus eros. Cras dapibus porttitor risus. Maecenas ultrices mi sed diam. Praesent gravida velit at elit vehicula porttitor. Phasellus nisl mi, sagittis ac, pulvinar id, gravida sit amet, erat. Vestibulum est. Lorem ipsum dolor sit amet, consectetur adipiscing elit. Curabitur id sem elementum leo rutrum hendrerit. Ut at mi. Donec tincidunt faucibus massa. Sed turpis quam, sollicitudin a, hendrerit eget, pretium ut, nisl. Duis hendrerit ligula. Nunc pulvinar congue urna.

4.1.2.2 Subsubsection Two

Nunc velit. Nullam elit sapien, eleifend eu, commodo nec, semper sit amet, elit. Nulla lectus risus, condimentum ut, laoreet eget, viverra nec, odio. Proin lobortis. Curabitur dictum arcu vel wisi. Cras id nulla venenatis tortor congue ultrices. Pellentesque eget pede. Sed eleifend sagittis elit. Nam sed tellus sit amet lectus ullamcorper tristique. Mauris enim sem, tristique eu, accumsan at, scelerisque vulputate, neque. Quisque lacus. Donec et ipsum sit amet elit nonummy aliquet. Sed viverra nisl at sem. Nam diam. Mauris ut dolor. Curabitur ornare tortor cursus velit.

4.1.2.2.1 Paragraph a Morbi tincidunt posuere arcu. Cras venenatis est vitae dolor. Vivamus scelerisque semper mi. Donec ipsum arcu, consequat scelerisque, viverra id, dictum at, metus. Lorem ipsum dolor sit amet, consectetur adipiscing elit. Ut pede sem, tempus ut, porttitor bibendum, molestie eu, elit. Suspendisse potenti. Sed id lectus sit amet purus faucibus vehicula. Praesent sed sem non dui pharetra interdum. Nam viverra ultrices magna.

4.1.2.2.2 Paragraph b Aenean laoreet aliquam orci. Nunc interdum elementum urna. Quisque erat. Nullam tempor neque. Maecenas velit nibh, scelerisque a, consequat ut, viverra in, enim. Duis magna. Donec odio neque, tristique et, tincidunt eu, rhoncus ac, nunc. Mauris malesuada malesuada elit. Etiam lacus mauris, pretium vel, blandit in, ultricies id, libero. Phasellus bibendum erat ut diam. In congue imperdiet lectus.

4.2 Section Two

Aenean scelerisque. Fusce pretium porttitor lorem. In hac habitasse platea dictumst. Nulla sit amet nisl at sapien egestas pretium. Nunc non tellus. Vivamus aliquet. Nam adipiscing euismod dolor. Aliquam erat volutpat. Nulla ut ipsum. Quisque tincidunt auctor augue. Nunc imperdiet ipsum eget elit. Aliquam quam leo, consectetur non, ornare sit amet, tristique quis, felis. Vestibulum ante ipsum primis in faucibus orci luctus et ultrices posuere cubilia Curae; Pellentesque interdum quam sit amet mi. Pellentesque mauris dui, dictum a, adipiscing ac, fermentum sit amet, lorem.

4.2.1 Subsection One

Ut quis wisi. Praesent quis massa. Vivamus egestas risus eget lacus. Nunc tincidunt, risus quis bibendum facilisis, lorem purus rutrum neque, nec porta tortor urna quis orci. Aenean aliquet, libero semper volutpat luctus, pede erat lacinia augue, quis rutrum sem ipsum sit amet pede. Vestibulum aliquet, nibh sed iaculis sagittis, odio dolor blandit augue, eget mollis urna tellus id tellus. Aenean aliquet aliquam nunc. Nulla ultricies justo eget orci. Phasellus tristique fermentum leo. Sed massa metus, sagittis ut, semper ut, pharetra vel, erat. Aliquam quam turpis, egestas vel, elementum in, egestas sit amet, lorem. Duis convallis, wisi sit amet mollis molestie, libero mauris porta dui, vitae aliquam arcu turpis ac sem. Aliquam aliquet dapibus metus.

4.2.2 Subsection Two

Vivamus commodo eros eleifend dui. Vestibulum in leo eu erat tristique mattis. Cras at elit. Cras pellentesque. Nullam id lacus sit amet libero aliquet hendrerit. Proin placerat, mi non elementum laoreet, eros elit tincidunt magna, a rhoncus sem arcu id odio. Nulla eget leo a leo egestas facilisis. Curabitur quis velit. Phasellus aliquam, tortor nec ornare rhoncus, purus urna posuere velit, et commodo risus tellus quis tellus. Vivamus leo turpis, tempus sit amet, tristique vitae, laoreet quis, odio. Proin scelerisque bibendum ipsum. Etiam nisl. Praesent vel dolor. Pellentesque vel magna. Curabitur urna. Vivamus congue urna in velit. Etiam ullamcorper elementum dui. Praesent non urna. Sed placerat quam non mi. Pellentesque diam magna, ultricies eget, ultrices placerat, adipiscing rutrum, sem.

4.2.2.1 Subsubsection One

Morbi sem. Nulla facilisi. Vestibulum ante ipsum primis in faucibus orci luctus et ultrices posuere cubilia Curae; Nulla facilisi. Morbi sagittis ultrices libero. Praesent eu ligula sed sapien auctor sagittis. Class aptent taciti sociosqu ad litora torquent per conubia nostra, per inceptos hymenaeos. Donec vel nunc. Nunc fermentum, lacus id aliquam porta, dui tortor

euismod eros, vel molestie ipsum purus eu lacus. Vivamus pede arcu, euismod ac, tempus id, pretium et, lacus. Curabitur sodales dapibus urna. Nunc eu sapien. Donec eget nunc a pede dictum pretium. Proin mauris. Vivamus luctus libero vel nibh.

4.2.2.2 Subsubsection Two

Fusce tristique risus id wisi. Integer molestie massa id sem. Vestibulum vel dolor. Pellentesque vel urna vel risus ultricies elementum. Quisque sapien urna, blandit nec, iaculis ac, viverra in, odio. In hac habitasse platea dictumst. Morbi neque lacus, convallis vitae, commodo ac, fermentum eu, velit. Sed in orci. In fringilla turpis non arcu. Donec in ante. Phasellus tempor feugiat velit. Aenean varius massa non turpis. Vestibulum ante ipsum primis in faucibus orci luctus et ultrices posuere cubilia Curae;

4.2.2.2.1 Paragraph a Aliquam tortor. Morbi ipsum massa, imperdiet non, consectetur vel, feugiat vel, lorem. Quisque eget lorem nec elit malesuada vestibulum. Quisque sollicitudin ipsum vel sem. Nulla enim. Proin nonummy felis vitae felis. Nullam pellentesque. Duis rutrum feugiat felis. Mauris vel pede sed libero tincidunt mollis. Phasellus sed urna rhoncus diam euismod bibendum. Phasellus sed nisl. Integer condimentum justo id orci iaculis varius. Quisque et lacus. Phasellus elementum, justo at dignissim auctor, wisi odio lobortis arcu, sed sollicitudin felis felis eu neque. Praesent at lacus.

4.2.2.2.2 Paragraph b Vivamus sit amet pede. Duis interdum, nunc eget rutrum dignissim, nisl diam luctus leo, et tincidunt velit nisl id tellus. In lorem tellus, aliquet vitae, porta in, aliquet sed, lectus. Phasellus sodales. Ut varius scelerisque erat. In vel nibh eu eros imperdiet rutrum. Donec ac odio nec neque vulputate suscipit. Nam nec magna. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Nullam porta, odio et sagittis iaculis, wisi neque fringilla sapien, vel commodo lorem lorem id elit. Ut sem lectus, scelerisque eget, placerat et, tincidunt scelerisque, ligula. Pellentesque non orci.

Evaluation and Discussion

Etiam vel ipsum. Morbi facilisis vestibulum nisl. Praesent cursus laoreet felis. Integer adipiscing pretium orci. Nulla facilisi. Quisque posuere bibendum purus. Nulla quam mauris, cursus eget, convallis ac, molestie non, enim. Aliquam congue. Quisque sagittis nonummy sapien. Proin molestie sem vitae urna. Maecenas lorem. Vivamus viverra consequat enim.

5.1 Section One

Nunc sed pede. Praesent vitae lectus. Praesent neque justo, vehicula eget, interdum id, facilisis et, nibh. Phasellus at purus et libero lacinia dictum. Fusce aliquet. Nulla eu ante placerat leo semper dictum. Mauris metus. Curabitur lobortis. Curabitur sollicitudin hendrerit nunc. Donec ultrices lacus id ipsum.

5.1.1 Subsection One

Donec a nibh ut elit vestibulum tristique. Integer at pede. Cras volutpat varius magna. Phasellus eu wisi. Praesent risus justo, lobortis eget, scelerisque ac, aliquet in, dolor. Proin id leo. Nunc iaculis, mi vitae accumsan commodo, neque sem lacinia nulla, quis vestibulum justo sem in eros. Quisque sed massa. Morbi lectus ipsum, vulputate a, mollis ut, accumsan placerat, tellus. Nullam in wisi. Vivamus eu ligula a nunc accumsan congue. Suspendisse ac libero. Aliquam erat volutpat. Donec augue. Nunc venenatis fringilla nibh. Fusce accumsan pulvinar justo. Nullam semper, dui ut dignissim auctor, orci libero fringilla massa, blandit pulvinar pede tortor id magna. Nunc adipiscing justo sed velit tincidunt fermentum.

5.1.2 Subsection Two

Integer placerat. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Sed in massa. Class aptent taciti sociosqu ad litora torquent per conubia nostra, per inceptos hymenaeos. Phasellus tempus aliquam risus. Aliquam rutrum purus at metus. Donec posuere odio at erat. Nam non nibh. Phasellus ligula. Quisque venenatis lectus in augue. Sed vestibulum dapibus neque.

5.1.2.1 Subsubsection One

Mauris tempus eros at nulla. Sed quis dui dignissim mauris pretium tincidunt. Mauris ac purus. Phasellus ac libero. Etiam dapibus iaculis nunc. In lectus wisi, elementum eu, sollicitudin nec, imperdiet quis, dui. Nulla viverra neque ac libero. Mauris urna leo, adipiscing eu, ultrices non, blandit eu, dui. Maecenas dui neque, suscipit sit amet, rutrum a, laoreet in, eros. Ut eu nibh. Fusce nec erat tempus urna fringilla tempus. Curabitur id enim. Sed ante. Cras sodales enim sit amet wisi. Nunc fermentum consequat quam.

5.1.2.2 Subsubsection Two

Ut auctor, augue porta dignissim vestibulum, arcu diam lobortis velit, vel scelerisque risus augue sagittis risus. Maecenas eu justo. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris congue ligula eget tortor. Nullam laoreet urna sed enim. Donec eget eros ut eros volutpat convallis. Praesent turpis. Integer mauris diam, elementum quis, egestas ac, rutrum vel, orci. Nulla facilisi. Quisque adipiscing, nulla vitae elementum porta, sem urna volutpat leo, sed porta enim risus sed massa. Integer ac enim quis diam sodales luctus. Ut eget eros a ligula commodo ultricies. Donec eu urna viverra dolor hendrerit feugiat. Aliquam ac orci vel eros congue pharetra. Quisque rhoncus, justo eu volutpat faucibus, augue leo posuere lacus, a rhoncus purus pede vel est. Proin ultrices enim.

5.1.2.2.1 Paragraph a Aenean tincidunt laoreet dui. Vestibulum ante ipsum primis in faucibus orci luctus et ultrices posuere cubilia Curae; Integer ipsum lectus, fermentum ac, malesuada in, eleifend ut, lorem. Vivamus ipsum turpis, elementum vel, hendrerit ut, semper at, metus. Vivamus sapien tortor, eleifend id, dapibus in, egestas et, pede. Pellentesque faucibus. Praesent lorem neque, dignissim in, facilisis nec, hendrerit vel, odio. Nam at diam ac neque aliquet viverra. Morbi dapibus ligula sagittis magna. In lobortis. Donec aliquet ultricies libero. Nunc dictum vulputate purus. Morbi varius. Lorem ipsum dolor sit amet, consectetur adipiscing elit. In tempor. Phasellus commodo porttitor magna. Curabitur vehicula odio vel dolor.

5.1.2.2.2 Paragraph b Praesent facilisis, augue a adipiscing venenatis, libero risus molestie odio, pulvinar consectetur felis erat ac mauris. Nam vestibulum rhoncus quam. Sed velit urna, pharetra eu, eleifend eu, viverra at, wisi. Maecenas ultrices nibh at turpis. Aenean quam. Nulla ipsum. Aliquam posuere luctus erat. Curabitur magna felis, lacinia et, tristique id, ultrices ut, mauris. Suspendisse feugiat. Cras eleifend wisi vitae tortor. Phasellus leo purus, mattis sit amet, auctor in, rutrum in, magna. In hac habitasse platea dictumst. Phasellus imperdiet metus in sem. Vestibulum ac enim non sem ultricies sagittis. Sed vel diam.

5.2 Section Two

Integer vel enim sed turpis adipiscing bibendum. Vestibulum pede dolor, laoreet nec, posuere in, nonummy in, sem. Donec imperdiet sapien placerat erat. Donec viverra. Aliquam eros. Nunc consequat massa id leo. Sed ullamcorper, lorem in sodales dapibus, risus metus sagittis lorem, non porttitor purus odio nec odio. Sed tincidunt posuere elit. Quisque eu enim. Donec libero risus, feugiat ac, dapibus eget, posuere a, felis. Quisque vel lectus ut metus tincidunt eleifend. Duis ut pede. Duis velit erat, venenatis vitae, vulputate a, pharetra sit amet, est. Etiam fringilla faucibus augue.

5.2.1 Subsection One

Aenean velit sem, viverra eu, tempus id, rutrum id, mi. Nullam nec nibh. Proin ullamcorper, dolor in cursus tristique, eros augue tempor nibh, at gravida diam wisi at purus. Donec mattis ullamcorper tellus. Phasellus vel nulla. Praesent interdum, eros in sodales sollicitudin, nunc nulla pulvinar justo, a euismod eros sem nec nibh. Nullam sagittis dapibus lectus. Nullam eget ipsum eu tortor lobortis sodales. Etiam purus leo, pretium nec, feugiat non, ullamcorper vel, nibh. Sed vel elit et quam accumsan facilisis. Nunc leo. Suspendisse faucibus lacus.

5.2.2 Subsection Two

Pellentesque interdum sapien sed nulla. Proin tincidunt. Aliquam volutpat est vel massa. Sed dolor lacus, imperdiet non, ornare non, commodo eu, neque. Integer pretium semper justo. Proin risus. Nullam id quam. Nam neque. Duis vitae wisi ullamcorper diam congue ultricies. Quisque ligula. Mauris vehicula.

5.2.2.1 Subsubsection One

Curabitur nunc magna, posuere eget, venenatis eu, vehicula ac, velit. Aenean ornare, massa a accumsan pulvinar, quam lorem laoreet purus, eu sodales magna risus molestie lorem. Nunc erat velit, hendrerit quis, malesuada ut, aliquam vitae, wisi. Sed posuere. Suspendisse ipsum arcu, scelerisque nec, aliquam eu, molestie tincidunt, justo. Phasellus iaculis. Sed posuere lorem non ipsum. Pellentesque dapibus. Suspendisse quam libero, laoreet a, tincidunt eget, consequat at, est. Nullam ut lectus non enim consequat facilisis. Mauris leo. Quisque pede ligula, auctor vel, pellentesque vel, posuere id, turpis. Cras ipsum sem, cursus et, facilisis ut, tempus euismod, quam. Suspendisse tristique dolor eu orci. Mauris mattis. Aenean semper. Vivamus tortor magna, facilisis id, varius mattis, hendrerit in, justo. Integer purus.

5.2.2.2 Subsubsection Two

Vivamus adipiscing. Curabitur imperdiet tempus turpis. Vivamus sapien dolor, congue venenatis, euismod eget, porta rhoncus, magna. Proin condimentum pretium enim. Fusce fringilla, libero et venenatis facilisis, eros enim cursus arcu, vitae facilisis odio augue vitae orci. Aliquam varius nibh ut odio. Sed condimentum condimentum nunc. Pellentesque eget massa. Pellentesque quis mauris. Donec ut ligula ac pede pulvinar lobortis. Pellentesque euismod. Class aptent taciti sociosqu ad litora torquent per conubia nostra, per inceptos hymenaeos. Praesent elit. Ut laoreet ornare est. Phasellus gravida vulputate nulla. Donec sit amet arcu ut sem tempor malesuada. Praesent hendrerit augue in urna. Proin enim ante, ornare vel, consequat ut, blandit in, justo. Donec felis elit, dignissim sed, sagittis ut, ullamcorper a, nulla. Aenean pharetra vulputate odio.

5.2.2.2.1 Paragraph a Quisque enim. Proin velit neque, tristique eu, eleifend eget, vestibulum nec, lacus. Vivamus odio. Duis odio urna, vehicula in, elementum aliquam, aliquet laoreet, tellus. Sed velit. Sed vel mi ac elit aliquet interdum. Etiam sapien neque, convallis et, aliquet vel, auctor non, arcu. Aliquam suscipit aliquam lectus. Proin tincidunt magna sed wisi. Integer blandit lacus ut lorem. Sed luctus justo sed enim.

5.2.2.2.2 Paragraph b Morbi malesuada hendrerit dui. Nunc mauris leo, dapibus sit amet, vestibulum et, commodo id, est. Pellentesque purus. Pellentesque tristique, nunc ac pulvinar adipiscing, justo eros consequat lectus, sit amet posuere lectus neque vel augue. Cras consectetur libero ac eros. Ut eget massa. Fusce sit amet enim eleifend sem dictum auctor. In eget risus luctus wisi convallis pulvinar. Vivamus sapien risus, tempor in, viverra in, aliquet pellentesque, eros. Aliquam euismod libero a sem.

General Conclusion and Perspectives

Nunc velit augue, scelerisque dignissim, lobortis et, aliquam in, risus. In eu eros. Vestibulum ante ipsum primis in faucibus orci luctus et ultrices posuere cubilia Curae; Curabitur vulputate elit viverra augue. Mauris fringilla, tortor sit amet malesuada mollis, sapien mi dapibus odio, ac imperdiet ligula enim eget nisl. Quisque vitae pede a pede aliquet suscipit. Phasellus tellus pede, viverra vestibulum, gravida id, laoreet in, justo. Cum sociis natoque penatibus et magnis dis parturient montes, nascetur ridiculus mus. Integer commodo luctus lectus. Mauris justo. Duis varius eros. Sed quam. Cras lacus eros, rutrum eget, varius quis, convallis iaculis, velit. Mauris imperdiet, metus at tristique venenatis, purus neque pellentesque mauris, a ultrices elit lacus nec tortor. Class aptent taciti sociosqu ad litora torquent per conubia nostra, per inceptos hymenaeos. Praesent malesuada. Nam lacus lectus, auctor sit amet, malesuada vel, elementum eget, metus. Duis neque pede, facilisis eget, egestas elementum, nonummy id, neque.

Proin non sem. Donec nec erat. Proin libero. Aliquam viverra arcu. Donec vitae purus. Donec felis mi, semper id, scelerisque porta, sollicitudin sed, turpis. Nulla in urna. Integer varius wisi non elit. Etiam nec sem. Mauris consequat, risus nec congue condimentum, ligula ligula suscipit urna, vitae porta odio erat quis sapien. Proin luctus leo id erat. Etiam massa metus, accumsan pellentesque, sagittis sit amet, venenatis nec, mauris. Praesent urna eros, ornare nec, vulputate eget, cursus sed, justo. Phasellus nec lorem. Nullam ligula ligula, mollis sit amet, faucibus vel, eleifend ac, dui. Aliquam erat volutpat.



Future Perspectives. Fusce vehicula, tortor et gravida porttitor, metus nibh congue lorem, ut tempus purus mauris a pede. Integer tincidunt orci sit amet turpis. Aenean a metus. Aliquam vestibulum lobortis felis. Donec gravida. Sed sed urna. Mauris et orci. Integer ultrices feugiat ligula. Sed dignissim nibh a massa. Donec orci dui, tempor sed, tincidunt nonummy, viverra sit amet, turpis. Quisque lobortis. Proin venenatis tortor nec wisi. Vestibulum placerat. In hac habitasse platea dictumst. Aliquam porta mi quis risus. Donec sagittis luctus diam. Nam ipsum elit, imperdiet vitae, faucibus nec, fringilla eget, leo. Etiam quis dolor in sapien porttitor imperdiet.

Glossary

Term One Definition of the first term goes here.

Term Two Definition of the second term goes here.

Term Three Definition of the third term goes here.

Acronyms

Acronym	Full Form
AI	Artificial Intelligence
ML	Machine Learning
API	Application Programming Interface
DL	Deep Learning
IoT	Internet of Things

Supplementary Material

Cras pretium. Nulla malesuada ipsum ut libero. Suspendisse gravida hendrerit tellus. Maecenas quis lacus. Morbi fringilla. Vestibulum odio turpis, tempor vitae, scelerisque a, dictum non, massa. Praesent erat felis, porta sit amet, condimentum sit amet, placerat et, turpis. Praesent placerat lacus a enim. Vestibulum non eros. Ut congue. Donec tristique varius tortor. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Nam dictum dictum urna.

Phasellus vestibulum orci vel mauris. Fusce quam leo, adipiscing ac, pulvinar eget, molestie sit amet, erat. Sed diam. Suspendisse eros leo, tempus eget, dapibus sit amet, tempus eu, arcu. Vestibulum wisi metus, dapibus vel, luctus sit amet, condimentum quis, leo. Suspendisse molestie. Duis in ante. Ut sodales sem sit amet mauris. Suspendisse ornare pretium orci. Fusce tristique enim eget mi. Vestibulum eros elit, gravida ac, pharetra sed, lobortis in, massa. Proin at dolor. Duis accumsan accumsan pede. Nullam blandit elit in magna lacinia hendrerit. Ut nonummy luctus eros. Fusce eget tortor.

Bibliography

- [1] John R. Jensen. *Introductory Digital Image Processing: A Remote Sensing Perspective*. 4th. Glenview, IL: Pearson Education, 2016.
- [2] Prasad S. Thenkabail and John G. Lyon. *Hyperspectral Remote Sensing of Vegetation*. 2nd. Boca Raton, FL: CRC Press, 2018.
- [3] M. Drusch et al. “Sentinel-2: ESA’s Optical High-Resolution Mission for GMES Operational Services”. In: *Remote Sensing of Environment* 120 (2012), pp. 25–36.
- [4] European Space Agency. *Sentinel-2 User Handbook*. Tech. rep. GMES-S2AS-DI-XXXX. ESA, 2015.
- [5] Microsoft. *Bing Maps Tile System*. <https://docs.microsoft.com/en-us/bingmaps/articles/bing-maps-tile-system>. Accessed: 2024. 2023.
- [6] Kirsten M. de Beurs and Geoffrey M. Henebry. *Land Surface Phenology*. Berlin, Germany: Springer, 2017.
- [7] Jordi Inglada et al. “Improved Early Crop Type Identification by Joint Use of High Temporal Resolution SAR and Optical Image Time Series”. In: *Remote Sensing* 8.5 (2016), p. 362.
- [8] Pierre Defourny et al. “Near Real-Time Agriculture Monitoring at National Scale at Parcel Resolution: Performance Assessment of the Sen2-Agri Automated System in Various Cropping Systems around the World”. In: *Remote Sensing of Environment* 221 (2019), pp. 551–568.
- [9] Ministry of Agriculture and Animal Resources. *Strategic Plan for the Transformation of Agriculture in Rwanda – Phase IV*. Tech. rep. Kigali, Rwanda: MINAGRI, 2018.
- [10] Giorgos Mountrakis, Jungho Im, and Caesar Ogole. “Support Vector Machines in Remote Sensing: A Review”. In: *ISPRS Journal of Photogrammetry and Remote Sensing* 66.3 (2011), pp. 247–259.
- [11] Leo Breiman. “Random Forests”. In: *Machine Learning* 45.1 (2001), pp. 5–32.
- [12] Tianqi Chen and Carlos Guestrin. “XGBoost: A Scalable Tree Boosting System”. In: *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*. 2016, pp. 785–794.
- [13] Yann LeCun, Yoshua Bengio, and Geoffrey Hinton. “Deep Learning”. In: *Nature* 521 (2015), pp. 436–444.

- [14] Emmanuel Maggiori et al. “Convolutional Neural Networks for Large-Scale Remote Sensing Image Classification”. In: *IEEE Transactions on Geoscience and Remote Sensing* 55.2 (2017), pp. 645–657.
- [15] Nataliia Kussul et al. “Deep Learning Classification of Land Cover and Crop Types Using Remote Sensing Data”. In: *IEEE Geoscience and Remote Sensing Letters* 14.5 (2017), pp. 778–782.
- [16] Victor Maus et al. “A Time-Weighted Dynamic Time Warping Method for Land-Use and Land-Cover Mapping”. In: *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing* 9.8 (2016), pp. 3729–3739.
- [17] Emanuele Barca et al. “Phenology-Based Crop Mapping Using Sentinel-2 Time Series”. In: *Remote Sensing* 12.14 (2020), p. 2227.
- [18] Sepp Hochreiter and Jürgen Schmidhuber. “Long Short-Term Memory”. In: *Neural Computation* 9.8 (1997), pp. 1735–1780.
- [19] Marc Rußwurm and Marco Körner. “Temporal Vegetation Modelling Using Long Short-Term Memory Networks for Crop Identification from Medium-Resolution Remote Sensing Data”. In: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition Workshops* (2017), pp. 11–19.
- [20] Roberto Interdonato et al. “DuPLO: A DUal view Point deep Learning architecture for time series classificatiOn”. In: *ISPRS Journal of Photogrammetry and Remote Sensing* 149 (2019), pp. 91–104.
- [21] Kyunghyun Cho et al. “Learning Phrase Representations using RNN Encoder-Decoder for Statistical Machine Translation”. In: *arXiv preprint arXiv:1406.1078* (2014).
- [22] Vivien Sainte Fare Garnot et al. “Satellite Image Time Series Classification with Pixel-Set Encoders and Temporal Self-Attention”. In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition* (2020), pp. 12325–12334.
- [23] Theoneste Mugiraneza, Andrea Nascetti, and Yifang Ban. “Continuous Monitoring of Urban Land Cover Change Trajectories with Landsat Time Series and LandTrendr-Google Earth Engine Cloud Computing”. In: *Remote Sensing* 12.18 (2020), p. 2883.